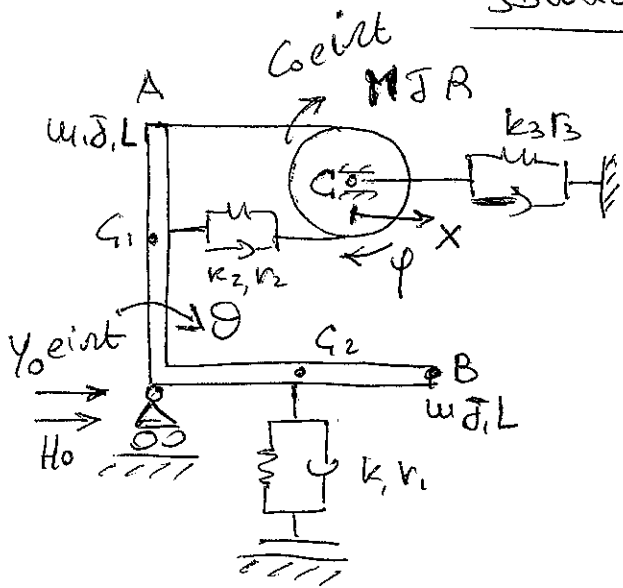


Soluzione 30 giugno 2008



$$\underline{x} = \begin{pmatrix} \theta \\ x \\ y \end{pmatrix}$$

$$E_c = \frac{1}{2} m \dot{x}_{g1}^2 + \frac{1}{2} m \dot{x}_{g2}^2 + \frac{1}{2} m \dot{y}_{g2}^2 + \frac{1}{2} I_1 \dot{\theta}^2 + \frac{1}{2} M \dot{x}^2 + \frac{1}{2} I \dot{\phi}^2$$

$$X_{S1} = \frac{L}{2}\theta + y \quad \varphi = \frac{L}{R}\theta + \frac{y}{R} - \frac{x}{R}$$

$$X_{s2} = y$$

$$\psi_{92} = -\frac{L}{2} \theta$$

$$[u_F] = \text{diag}([u \quad u \quad u \quad 2\bar{J}, \quad M \quad \bar{D}])$$

$$(\tilde{J}_m) = \begin{bmatrix} L/2 & 0 & 1 \\ 0 & 0 & 1 \\ -L/2 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ L/2 & -1/2 & 1/2 \end{bmatrix}$$

$$V_{EL} = \frac{1}{2} k_1 \Delta l_1^2 + \frac{1}{2} k_2 \Delta l_2^2 + \frac{1}{2} k_3 \Delta l_3^2$$

$$(\mathcal{J}_{EL}) = \begin{bmatrix} -L/2 & 0 & 0 \\ -2(L-R) & +2 & -2 \\ 0 & -1 & 0 \end{bmatrix}$$

$$(k_{\text{eff}}) = \text{diag}(k_1, k_2, k_3)$$

$$\Delta L_1 = -\frac{L}{2} \theta$$

$$\Delta L_2 = -\frac{L}{2(L-r)} \theta - 2\gamma + 2x$$

$$\Delta b_3 = -x$$

$$V_A = mg h_{A1} + mg h_{A2} =$$

$$= mg \frac{L}{2} \cos \theta + mg \frac{L}{2} \sin \theta$$

$$\frac{\partial^2 V_g}{\partial \theta^2} = -mg \frac{L}{2} \cos \theta + mg \frac{L}{2} \sin \theta$$

$$\frac{\partial^2 V_G}{\partial \theta^2} \Big|_0 = -mg \frac{L}{2} \rightarrow V_{G2} = \frac{1}{2} \left(-mg \frac{L}{2} \right) \theta^2 \rightarrow$$

$$\rightarrow [K_9] = \begin{bmatrix} -\omega g L/2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

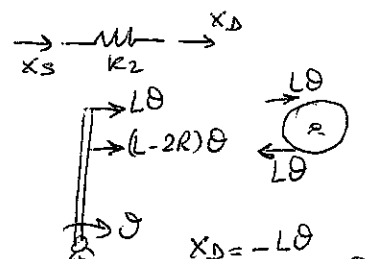
$$\delta W = C_0 e^{i\omega t} \delta\phi + H_0 \delta\psi = C_0 e^{i\omega t} \left(\frac{L}{R} \delta\theta + \frac{\delta\psi}{R} - \frac{\delta x}{R} \right) + H_0 \delta\psi$$

$$Q_{LC} = \left\{ \frac{L}{R} - \frac{1}{R} \right\}^T C_{\text{eint}}$$

$$Q_{VC} = \frac{C_0}{\beta} e^{i\omega t} \quad ; \quad Q_{VH} = H_0$$

calcolo di ΔI_2 con
 $x=y=0$; $\theta \neq 0$:

$$\Delta L_2 = x_D - x_S$$



$$X_D = -L\theta$$

$$X_S = +(L-2R)\theta$$

$$\Delta L_2 = -2(L-R)\theta$$